

Clodomir Santana

Postdoctoral Scholar · Division of Cardiovascular Medicine, Department of Internal Medicine · UC Davis Health

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PROFILE

Postdoctoral scientist bridging computer science and cardiovascular medicine, with a focus on harmonizing, integrating, and analyzing large-scale clinical and research datasets. Current work applies machine learning, geospatial analysis, and survival modeling to heart failure with preserved ejection fraction (HFpEF), a heterogeneous syndrome that disproportionately affects women and has limited effective therapies. As a Fellow of the NIH-funded HeartShare consortium, develops methods to phenotype and analyze HFpEF cohorts across multiple clinical trials and observational studies. As fellow of the AHA-funded SFRN UC Davis PRECISE Center, integrates wearable biosignal, psychosocial survey data, and place-based environmental data into unified analytic frameworks. Given the disproportionate burden of HFpEF and other chronic conditions on women's health, interested in applying computational modeling and new approach methodologies to study hormonal regulation and sex differences in cardiovascular disease.

EDUCATION

Ph.D., Computer Science — University of Exeter, Exeter, UK 2019 – 2023

Doctoral scholarship, College of Engineering, Mathematics and Physical Sciences

M.Sc., Systems Engineering — University of Pernambuco, Recife, Brazil 2017 – 2019

Honorable Mention, Outstanding Master's Dissertation (UPE)

Visiting International Student, Computer Engineering — Memorial University, St. John's, Canada 2013 – 2014

Brazilian Scientific Mobility Program (Science Without Borders, CAPES)

B.Sc., Computer Engineering — University of Pernambuco, Recife, Brazil 2010 – 2016

RESEARCH EXPERIENCE

Postdoctoral Scholar — University of California, Davis (UC Davis Health) 2024 – Present

Division of Cardiovascular Medicine (Dr. Martin Cadeiras) & AI for Health Center (Dr. Vladimir Filkov)

- **Cross-trial data harmonization (HeartShare consortium):** harmonize and phenotype HFpEF patients across multiple independent clinical-trial cohorts (TOPCAT, RELAX, NEAT-HFpEF, INDIE-HFpEF), reconciling heterogeneous variable definitions across studies, and develop individualized treatment-effect models that identify which patients benefit from therapy.
- **Geospatial and exposomic data integration:** designed and led a pipeline integrating built-environment and neighborhood-level exposures at high spatial resolution across California; identified geographic HFpEF clusters ($N = 6,702$) with marked survival differences (adjusted HR ≈ 1.25).
- **Multimodal clinical data integration (PRECISE / SFRN center):** integrate wearable-derived physiological signals, psychosocial survey instruments, and clinical cohort data, including circadian-rhythm analysis and standardized cardiovascular-health scoring.
- **Clinical and biomedical NLP:** apply natural language processing to biomedical literature to map and prioritize protein–disease associations in HFpEF.
- Collaborate with clinical cardiologists to align computational methods with clinical research questions; mentor data-science trainees and supervise student collaborators.

Postdoctoral Researcher, Digital Humanities — Institute of History, Polish Academy of Sciences

2023 – 2024

Warsaw, Poland

- Applied NLP, large language models, and data-mining methods to large historical corpora; built **structured, standardized datasets and authority dictionaries** of people, places, and institutions, resolving duplicate records, name variants, and entity changes over time.
- Examined the role of rigorous data preprocessing and transparent documentation in ensuring result quality and reproducibility across data sources.

AI Scientist (KTP Associate) — City Science, University of Exeter

2022 – 2023

Faculty of Environment, Science and Economy, Exeter, UK

- Developed AI and network-science models for urban systems, including human-mobility analysis during the COVID-19 pandemic (partnership with Cuebiq Inc.) and analysis of targeted policing interventions (Devon & Cornwall Police).

Artificial Intelligence Engineer / R&D — PickCells

2019

Recife, Brazil

- Built deep-learning methods for automated parasitological diagnostics and disease-vector image analysis.

Software Engineer — Motorola Mobility

2015 – 2017

Recife, Brazil

- Software engineering across analyst and engineering roles (internship to full engineer).

TECHNICAL SKILLS

- **Programming & data:** Python, R, SQL, Git; reproducible analysis pipelines (Jupyter).
- **Data harmonization & integration:** multi-cohort and multi-trial harmonization, common data elements, clinical-trial and EHR data, wearable biosignals, multi-omics integration, entity resolution and standardization.
- **Machine learning & statistics:** supervised and unsupervised learning, ensemble methods (Super Learner), clustering and phenotyping, survival analysis, individualized treatment-effect and causal-inference methods.
- **Geospatial:** GIS, spatial clustering, built-environment / exposomic feature engineering (OpenStreetMap, area-based deprivation indices).
- **NLP:** clinical and scientific text mining, named-entity resolution, large language models.
- **Languages:** English (fluent), Portuguese (native).

SELECTED PUBLICATIONS

Cardiovascular & Health Data Science

Santana C, Mukherjee C, Quazi A, Menezes R, Filkov V, Sigdel D, et al. Natural language processing of biomedical text to map and prioritize protein–disease associations in HFpEF. *Computers in Biology and Medicine*. 2026;205:111599.

Santana C, Cadeiras M, Liem D. Multi-organ crosstalk in HFpEF: reframing a cardiac syndrome as a widespread systemic disease. *The Journal of Precision Medicine: Health and Disease*. 2026;100042.

Balabhadruni, M, Rocha, P, **Santana, C**, Liem, D and Cadeiras, M. Assessing heart transplant inequities across California's neighborhood environments. *Journal of the American College of Cardiology*. 2026. (Abstract)

El Mekkaoui K, Mukherjee C, **Santana C**, Silva H, Quazi A, Balabhadruni M, et al. The impact of the social exposome in cardiovascular health and disease. *The Journal of Precision Medicine: Health and Disease*. 2025;3:100015.

Santana C, Botta F, Barbosa H, Privitera F, Menezes R, Di Clemente R. COVID-19 is linked to changes in the time–space dimension of human mobility. *Nature Human Behaviour*. 2023;7(10):1729–1739.

Caetano A, **Santana C**, de Lima RA. Diagnostic support of parasitic infections with an AI-powered microscope. *Research on Biomedical Engineering*. 2023;39(3):561–572.

de Santana CJ, Firmo ACA, de Oliveira RFAP, Lins PJB, de Lima GA, de Lima RA. A solution for counting *Aedes aegypti* and *Aedes albopictus* eggs in paddles from ovitraps using deep learning. *IEEE Latin America Transactions*. 2019;17(12):1987–1994.

Machine Learning, Network Science & Data Engineering

Santana, C, Vasques Filho, D., Bojanowski, M., & Bloch, A. Unveiling the critical nexus of data preprocessing and transparent documentation for result quality and reproducibility in digital history. *Digital Humanities Quarterly*. 2025.

Santana C, Oliveira M, Bastos-Filho C, Menezes R. Beyond exploitation: measuring the impact of local search in swarm-based memetic algorithms through the interactions of individuals in the population. *Swarm and Evolutionary Computation*. 2022;70:101040.

Santana C, Macedo M, Alves E, Guerreiro MT, Siqueira HV, Gokhale A, Bastos-Filho C. Bio-inspired multi-objective algorithms applied on the optimization of the AODV's routing recovery mechanism. *IEEE Access*. 2023;11:116480–116496.

Macedo M, Siqueira H, Figueiredo E, **Santana C**, Lira RC, Gokhale A, Bastos-Filho C. Overview on binary optimization using swarm-inspired algorithms. *IEEE Access*. 2021;9:149814–149858.

Siqueira H, **Santana C**, Macedo M, Figueiredo E, Gokhale A, Bastos-Filho C. Simplified binary cat swarm optimization. *Integrated Computer-Aided Engineering*. 2021;28(1):35–50.

Santana Jr CJ, Macedo M, Siqueira H, Gokhale A, Bastos-Filho CJ. A novel binary artificial bee colony algorithm. *Future Generation Computer Systems*. 2019;98:180–196.

Figueiredo E, Macedo M, Siqueira HV, **Santana Jr CJ**, Gokhale A, Bastos-Filho CJ. Swarm intelligence for clustering — a systematic review with new perspectives on data mining. *Engineering Applications of Artificial Intelligence*. 2019;82:313–329.

Selected Conference Proceedings & Book Chapters

Santana C, Keedwell E, Menezes R. An approach to assess swarm intelligence algorithms based on complex networks. *Proc. Genetic and Evolutionary Computation Conference (GECCO)*. 2020:31–39.

Santana C, Keedwell E, Menezes R. Networks of evolution: modelling and deconstructing genetic algorithms using dynamic networks. *Proc. GECCO Companion*. 2022:459–462.

Bastos-Filho CJA, de Lima-Neto FB, Lins AJDCC, de Lacerda MGP, Macedo MGM, **de Santana Junior CJ**, et al. Fish School Search: account for the first decade. In: *Handbook of AI-based Metaheuristics*. CRC Press; 2021:21–42.

Guimarães F, Bastos-Filho C, **Santana C**. Impact of local search in the memetic particle swarm optimization. In: *Handbook of Formal Optimization*. Springer; 2023:1–20.

Alencar RC, **Santana C**, Bastos-Filho CJ. Optimizing routes for medicine distribution using team ant colony system. *Int. Conf. on Hybrid Intelligent Systems*. Springer; 2018:40–49.

Santana C, Bastos-Filho CJ, Macedo M, Siqueira H. SBFSS: simplified binary fish school search. *IEEE Congress on Evolutionary Computation (CEC)*. 2019:2595–2602.

A full publication list (30+ items, including additional peer-reviewed conference papers in clustering, optimization, and network science) is available on request.

HONORS & AWARDS

- HeartShare Research Skills Fellowship, HeartShare Consortium (2025).
- Honorable Mention for Outstanding Master's Dissertation, University of Pernambuco (2019).
- Ph.D. Research Scholarship, University of Exeter (2019–2023).
- Scientific Scholarship, Foundation for the Support of Science and Technology of the State of Pernambuco (FACEPE, 2017–2019).
- Scientific Mobility Program (Science Without Borders), CAPES, Brazil (2013–2014).

PROFESSIONAL SERVICE — PEER REVIEW

Swarm and Evolutionary Computation · Information Sciences · Applied Soft Computing · Soft Computing · International Journal of Fuzzy Systems · IEEE Latin America Transactions · Applied Network Science · EPJ Data Science · International Conference on Complex Networks · Frontiers in Computer Science · Frontiers in Systems Biology · SN Computer Science · Frontiers in Public Health.